SOLUTIONS ENGINEER TECHNICAL IMMERSION: 6-WEEK ROADMAP

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THE UPDATED 6-WEEK ARC

Week	Technical Focus	SE Sales Focus	Interview Story
Week 1	SQL Mastery	Discovery Through Data	“How I diagnose customer problems”
Week 2	Python + API Fundamentals	Technical Troubleshooting	“How I debug customer integration issues”
Week 3	AWS Solutions Architect	Pre-Sales Methodology	“How I architect for the customer”
Week 4	GCP (3 days) + SE Sales Deep Dive	Discovery, Demo, POC, Close	“My end-to-end SE sales cycle story”
Week 5	LLM Fundamentals + RAG	AI Deployment Strategy	“How I guide AI adoption for customers”
Week 6	AI Agents + Proof of Concept Mastery	The SE POC That Wins Deals	“The AI POC I built for a customer”

WEEK 1: SQL MASTERY + DISCOVERY THROUGH DATA

Technical Focus

Master SQL so fluently that you can write a query that answers any business question.

Resources

- **Udemy:** “The Complete SQL Bootcamp: Go from Zero to Hero” by Jose Portilla
- **O’Reilly:** “Getting Started with SQL” by Thomas Nield
- **Free:** SQLBolt.com, Mode Analytics SQL Tutorial, HackerRank SQL

SE Sales Application (1 hour/day)

Every SE discovery workshop ends with data questions. This week teaches you to answer them.

- **Monday:** Read “The 5 Whys” discovery framework. Practice asking customers questions that lead to data-diagnosable problems.
- **Tuesday:** Watch 3 YouTube videos of real SE discovery calls (search “solutions engineer discovery call”). Notice when the SE uses “show me the data.”
- **Wednesday:** Map the BANT and MEDDPICC qualification frameworks. Understand how SEs support AE qualification.
- **Thursday:** Study how Snowflake SEs position their platform. Watch their public YouTube demos.
- **Friday:** Write out a 10-question discovery script for a mock healthcare customer.
- **Saturday:** Practice running a mock discovery call with yourself recorded.

Interview Story Building (1 hour/day)

Every SE interview asks behavioral questions. This week builds your “data diagnosis” story.

Build ONE story this week using the STAR framework (Situation, Task, Action, Result):

Story 1: “Tell me about a time you diagnosed a customer problem using data.” - Situation: Your Cigna work — employer groups weren’t completing onboarding - Task: Figure out why completion rates were low across 13 territories - Action: Conducted root cause analysis by geography, identified that same-day availability mattered more than proximity - Result: 25% lift in booking completion after change management

Practice telling this story in under 90 seconds. Record yourself. Refine it daily until it flows naturally.

Week 1 Deliverable

- SQL fluency (can write SELECT, JOIN, GROUP BY, subqueries from memory)
- GitHub Project 1: Healthcare Booking Analysis (your Cigna story turned into a database + queries)

- One rehearsed 90-second story about diagnosing a customer problem
- LinkedIn post showing the database you built

WEEK 2: PYTHON + API FUNDAMENTALS + TECHNICAL TROUBLESHOOTING

Technical Focus

Write Python scripts that call APIs, parse responses, and debug errors. This is the single highest-leverage technical skill for any SE role in 2026.

Resources

- **Udemy:** “100 Days of Code: The Complete Python Pro Bootcamp” by Angela Yu (Days 1-14)
- **O’Reilly:** “Python Crash Course, 3rd Edition” by Eric Matthes
- **Critical tool:** Postman (free download)
- **APIs to practice with:** OpenAI, Anthropic Claude, Shopify, JSONPlaceholder

SE Sales Application (1 hour/day)

Every SE role tests API debugging. This week builds the mental model of how customers actually fail.

- **Monday:** Read API documentation for 3 major platforms (Stripe, Twilio, Shopify). Notice the patterns.
- **Tuesday:** Study common API error codes (400, 401, 403, 404, 429, 500). Know what each means and what a customer would do wrong to trigger them.
- **Wednesday:** Watch “Postman Masterclass” on YouTube. Learn the workflow SEs actually use when a customer reports an integration issue.
- **Thursday:** Study how to read logs. Watch a video on Splunk or CloudWatch log analysis for error patterns.
- **Friday:** Practice the SE troubleshooting workflow: request → response → headers → status code → logs → reproduce → escalate or fix.
- **Saturday:** Map the difference between “customer error,” “integration error,” and “platform error.” Know when to escalate.

Interview Story Building (1 hour/day)

Story 2: “Tell me about a time you debugged a complex technical integration.” - Situation: Your Meta work — partners integrating the platform hit API errors - Task: Debug the integration without breaking production for 20M customers - Action: Analyzed request/response patterns, identified the root cause in authentication headers, partnered with engineering to fix - Result: Zero customer-impacting incidents during a 20M-user migration

Practice this story. Make it concrete. Know your numbers.

Week 2 Deliverable

- Python API fluency (can call OpenAI or Shopify API, parse JSON, handle errors)
- GitHub Project 2: AI Startup Readiness Chatbot (Python + OpenAI API + error handling)
- Rehearsed technical debugging story
- LinkedIn post: “What debugging taught me about the SE role”

WEEK 3: AWS SOLUTIONS ARCHITECT + PRE-SALES METHODOLOGY

Technical Focus

AWS is the universal SE cloud credential. Master the core services, then understand when and why to recommend them.

Resources

- **Udemy (MORNING):** “Ultimate AWS Certified Solutions Architect Associate 2026” by Stephane Maarek
- **O’Reilly (AFTERNOON):** “Mastering Technical Sales” by John Care (chapters 1-10)
- **O’Reilly:** “Great Demo!” by Peter Cohan
- **Free:** AWS Free Tier, AWS Skill Builder

Split Day Structure This Week

- **Mornings (4 hours):** AWS technical learning
- **Afternoons (4 hours):** SE Sales methodology + practice

SE Sales Application (1 hour/day) - EXPANDED THIS WEEK

This is the critical week where technical meets sales motion:

- **Monday:** MEDDPICC deep dive. Learn each letter. Practice applying to your own career story.
- **Tuesday:** The 3-Act Demo Structure (Great Demo! methodology). Situation → Solution → Support.
- **Wednesday:** Discovery Question Library — build 20 discovery questions across technical, business, and timeline dimensions.
- **Thursday:** Objection Handling — study how SEs handle “too expensive,” “we already use [competitor],” “we want to build in-house.”
- **Friday:** The Technical Win — when do you have it, how do you get it, what does “closing technical” mean?
- **Saturday:** Mock demo day. Record yourself doing a full 20-minute SE demo of an AWS-based solution.

Interview Story Building (1 hour/day)

Story 3: “Walk me through how you would architect a solution for a customer.” - Situation: A startup customer (use your Eventbrite work) wanted to scale their platform - Task: Architect a solution that met their needs within budget - Action: Conducted discovery, scoped use cases, recommended cloud-native architecture, presented to CTO - Result: Customer adopted the solution, 17% activation improvement

Week 3 Deliverable

- AWS core services fluency (EC2, S3, VPC, RDS, Lambda, IAM)
- Can sketch a cloud architecture diagram for a customer use case
- “Great Demo!” methodology internalized
- Rehearsed “how I architect” story
- LinkedIn post about the SE methodology you learned

WEEK 4: GCP (3 DAYS) + SE SALES MASTERY (3 DAYS)

Structure Change

First 3 days: Focused GCP learning for your Google Customer SE pipeline Last 3 days: Deep dive into full SE sales motion

Part 1: GCP (Monday-Wednesday)

Resources: - **Udemy:** "GCP Associate Cloud Engineer" by Ankit Mistry (focus on the sections that parallel your AWS knowledge) - **Free:** Google Cloud Skills Boost (free labs using real GCP console) - **Google Merchant API documentation** (directly relevant to your Google interview)

Daily Focus: - **Monday:** GCP fundamentals compared to AWS (you already know the concepts, just learn GCP's versions) - **Tuesday:** Google Cloud compute, storage, networking (what matters for your interview) - **Wednesday:** Google Merchant API deep dive + IAM (directly relevant to the role)

Part 2: SE Sales Mastery (Thursday-Saturday)

Resources: - **O'Reilly:** "The Qualified Sales Leader" by John McMahon - **O'Reilly:** Revisit "Mastering Technical Sales" chapters on closing - **YouTube:** PreSales Collective entire channel

Daily Focus: - **Thursday:** The full SE sales cycle — Discovery → Demo → POC → Technical Win → Close → Handoff - **Friday:** Managing the POC (Proof of Concept) — scope, timeline, success criteria, customer engagement - **Saturday:** The Executive Briefing — how SEs present to C-suite, the VRA framework (Value, Risk, Action)

Interview Story Building (1 hour/day)

Story 4: "Describe your end-to-end sales cycle experience." - Walk through a full SE motion from first customer call to closed deal - Use your Meta work — enterprise partner integrations from discovery to deployment - Include specific numbers: discovery calls, POC timeline, adoption metrics

Week 4 Deliverable

- GCP fluency for Google interview (Merchant API, IAM, core services)
- Full SE sales cycle mastery — can articulate every stage
- Rehearsed end-to-end sales cycle story
- GitHub Project 3: Multi-Cloud Architecture Comparison (AWS vs GCP for a customer scenario)

WEEK 5: LLM FUNDAMENTALS + RAG + AI DEPLOYMENT STRATEGY

Technical Focus

Master the AI stack that every modern SE interview tests. RAG is the single most-asked AI SE interview topic in 2026.

Resources

Udemy: - "LangChain - Develop LLM powered applications with LangChain" — the hands-on course - Build 7 practical apps including AI PDF reader, chatbot, personal assistant

O'Reilly (Priority Reading): - **"AI Engineering" by Chip Huyen** — the single most important book on your list - **"Hands-On RAG for Production"** by Mendelevitch & Bao - **"Learning LangChain"**

Free Tools: - OpenAI API ($5 free credits) - Anthropic Claude API (free tier) - LangChain (open-source) - Google Colab (free compute) - LangSmith (free observability)

Daily Focus

- **Monday:** LLM fundamentals + prompt engineering
- **Tuesday:** RAG — Retrieval Augmented Generation (the #1 AI SE topic)
- **Wednesday:** Vector databases + embeddings
- **Thursday:** LLM evaluation (hallucinations, faithfulness, relevance metrics)
- **Friday:** Observability + cost optimization for AI systems
- **Saturday:** Build "Chat with Docs" RAG agent — load your Forbes articles, query them

SE Sales Application (1 hour/day)

This week: How do you SELL AI to enterprise customers?

- **Monday:** Study OpenAI's enterprise case studies. Notice how they talk about ROI.
- **Tuesday:** Read Anthropic's customer stories. Notice the specific metrics they use.
- **Wednesday:** Study Databricks' LLM positioning for Lakehouse customers.
- **Thursday:** Learn to answer "Build vs Buy vs Fine-tune" — this is THE enterprise AI question.
- **Friday:** The AI ROI conversation — how do you quantify AI value for a CFO?
- **Saturday:** The AI risk conversation — hallucinations, bias, security, compliance. How does an SE reassure enterprise buyers?

Interview Story Building

Story 5: "Tell me how you'd guide an enterprise customer through their first AI deployment." - Draw from your Forbes AI content work + Eventbrite AI-native prototypes
- Walk through the full motion: AI readiness assessment, use case scoping, POC, evaluation, deployment

Week 5 Deliverable

- LLM and RAG fluency
- Can build a working RAG agent from scratch
- GitHub Project 4: Chat with Docs RAG Agent (your Forbes articles as the knowledge base)
- Rehearsed AI deployment story
- LinkedIn post: "What I learned about RAG this week"

WEEK 6: AI AGENTS + PROOF OF CONCEPT MASTERY

Technical Focus

AI Agents are the 2026 must-have skill. But Week 6's bigger focus is **the SE Proof of Concept** — how to build one, scope it, demo it, and close it.

Resources

Udemy: - "Complete Agentic AI Bootcamp with LangGraph and Langchain" - Focus on multi-agent systems, tool use, MCP (Model Context Protocol)

O'Reilly: - **"Building LLMs for Production"** — chapters on agents, fine-tuning, deployment - **"Hands-On LLM Serving and Optimization"** — enterprise deployment - **"Great Demo!"** — revisit for the POC chapters

Critical — Anthropic MCP: - Read the Model Context Protocol documentation at anthropic.com - This is what every SE at Anthropic must know

Daily Focus

- **Monday:** Agent fundamentals + tool use
- **Tuesday:** Multi-agent orchestration with LangGraph + CrewAI
- **Wednesday:** Model Context Protocol (MCP) — Anthropic's standard
- **Thursday:** Deploying AI agents to production (Vercel, Replit, Streamlit)
- **Friday:** The SE Proof of Concept workflow (this is the sales motion part)
- **Saturday:** BUILD YOUR SIGNATURE DEMO — "The Enterprise AI Readiness Assessment Agent"

SE Sales Application — WEEK-LONG EMPHASIS

The POC Mastery Week

This is the week where you learn to not just BUILD AI, but SELL IT through a Proof of Concept:

- **Monday:** POC Scoping — how do you define success criteria with a customer before writing any code?
- **Tuesday:** POC Timeline — how do you keep a POC from dragging into a free consulting engagement?
- **Wednesday:** POC Demo Day — how do you present POC results to get the deal signed?
- **Thursday:** POC Transition — how does the POC become production and a signed contract?
- **Friday:** The Value Realization Plan (VRA) — how SEs help customers measure ROI after adoption
- **Saturday:** Record your 15-minute capstone demo — this is your interview artifact

Interview Story Building — YOUR STAR STORY LIBRARY

By end of Week 6, you should have 6 fully rehearsed stories:

1. Data Diagnosis (Cigna root cause analysis story)
2. Technical Debugging (Meta API integration story)
3. Customer Architecture (Eventbrite solution architecture story)
4. End-to-End Sales Cycle (Meta enterprise deal story)
5. AI Deployment Guidance (Forbes + Eventbrite AI story)
6. The Proof of Concept (your capstone demo story — “I built an AI agent that…”)

Week 6 Deliverable

- GitHub Projects 5 and 6 — RAG Agent + Enterprise AI Readiness Agent
- Deployed live demo at a public URL
- 15-minute recorded demo walkthrough
- 6 rehearsed STAR stories
- Complete SE POC methodology
- LinkedIn profile updated with “AI Agent Builder”

YOUR GITHUB PORTFOLIO (6 Projects Across 6 Weeks)

#	Project	Week	Why It Matters for SE Interviews
1	Healthcare Booking Analysis	Week 1	Proves SQL + data diagnosis
2	AI Startup Readiness Chatbot	Week 2	Proves Python + API calling
3	Multi-Cloud Architecture (AWS + GCP)	Week 4	Proves cloud design + customer thinking
4	Chat with Docs RAG Agent	Week 5	Proves RAG — the top AI SE skill
5	Multi-Agent Workflow	Week 6	Proves agent orchestration
6	Enterprise AI Readiness Agent	Week 6	Your signature demo piece

YOUR INTERVIEW READINESS FRAMEWORK

The 6 Behavioral Stories Every SE Interview Tests

By end of Week 6, each of these should be a 90-second polished story:

1. **"How do you approach customer discovery?"** → Cigna 500+ workshops story
2. **"Tell me about a complex technical debugging situation."** → Meta API migration story
3. **"How do you architect a solution for a customer?"** → Eventbrite deployment story
4. **"Walk me through your sales cycle experience."** → Meta enterprise close story
5. **"How would you guide an AI deployment for an enterprise?"** → Forbes + Eventbrite AI story
6. **"Show me something you built recently."** → Your capstone AI agent demo

Technical Interview Topics You'll Own

- SQL — can write JOINs, GROUP BY, subqueries from memory
- Python — can call APIs, parse JSON, handle errors, build small applications
- Cloud — can explain AWS core services, compare GCP to AWS for customers
- AI Stack — can explain LLMs, RAG, agents, evaluation, observability
- Architecture — can draw a cloud architecture diagram for a customer scenario
- POC — can scope, deliver, and close a Proof of Concept

O'REILLY READING LIST (Final Priority Order)

#	Book	Week	Why
1	"Getting Started with SQL" — Thomas Nield	Week 1	Practical, short, excellent intro
2	"Python Crash Course" — Eric Matthes	Week 2	Best Python intro for non-developers
3	"Mastering Technical Sales" — John Care	Week 3	**THE SE BIBLE**
4	"Great Demo!" — Peter Cohan	Week 3	Demo methodology every SE uses
5	"The Qualified Sales Leader" — John McMahon	Week 4	AE-SE partnership understanding
6	"AI Engineering" — Chip Huyen	Week 5	**Most important AI book in 2025**
7	"Hands-On RAG for Production"	Week 5	Enterprise RAG mastery
8	"Learning LangChain"	Week 5	Framework fluency
9	"Building LLMs for Production"	Week 6	Agent + deployment depth
10	"Hands-On LLM Serving and Optimization"	Week 6	Enterprise deployment

YOUR WEEKLY LINKEDIN POSTS (One Per Week During Immersion)

Stay visible without applying:

- **Week 1:** “Built a healthcare data analysis project. Here’s what SQL taught me about discovery.”
- **Week 2:** “Debugged an API integration from scratch. Here’s why every SE needs Python + Postman.”
- **Week 3:** “Went deep on AWS this week. Here’s what every SE should know about cloud architecture conversations with customers.”
- **Week 4:** “Studied GCP vs AWS this week. Here’s the framework I’d use to guide customers choosing between them.”
- **Week 5:** “Built my first RAG agent. Here’s why enterprise AI is the highest-value SE skill in 2026.”
- **Week 6:** “6 weeks ago I committed to closing my technical gaps. Today I have 6 GitHub projects, deployed AI agents, and the confidence to demo live in any interview. Here’s what I learned.”

CRITICAL INSIGHT ON SE SALES VS TECHNICAL

The ratio of technical to sales in your immersion should match the ratio in the actual role:

- **An SE is 60% technical, 40% sales skill.** Your immersion reflects this — 32 hours/week technical, 12 hours/week SE sales application, 4 hours/week interview story building.
- **Technical skills get you interviews. Sales skills get you offers.** A candidate who can demo an AI agent AND articulate the MEDDPICC framework beats a candidate who can only do one.
- **Every technical skill needs a customer conversation attached to it.** That's why each week has the "SE Sales Application" block — you're not just learning SQL, you're learning how to use SQL in a customer discovery call.

THE BIG PICTURE

You asked for something critical: not just learning but applying. This plan makes every day a learning-AND-applying day. You'll finish 6 weeks with:

- 6 production-ready GitHub projects
- 6 rehearsed behavioral stories
- 3 certifications scheduled (Google Associate Cloud Engineer, AWS Solutions Architect Associate, Databricks Data Engineer Associate)
- Complete SE sales methodology internalized
- LLM + AI Agent fluency
- A live-deployed AI demo
- Community connections at LangChain, OpenAI, Anthropic, Databricks
- An identity shift from "PM transitioning to SE" to "SE who builds and sells AI"

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